

Towards Cross-Tokenizer Distillation: the Universal Logit Distillation Loss for LLMs

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Abstract

Deploying large language models (LLMs) of several billion parameters can be impractical in most industrial use cases due to constraints such as cost, latency limitations, and hardware accessibility. Knowledge distillation (KD) offers a solution by compressing knowledge from resource-intensive large models to smaller ones. Various strategies exist, some relying on the text generated by the teacher model and optionally utilizing his logits to enhance learning. However, these methods based on logits often require both teacher and student models to share the same tokenizer, limiting their applicability across different LLM families. In this paper, we introduce Universal Logit Distillation (ULD) loss, grounded in optimal transport, to address this limitation. Our experimental results demonstrate the effectiveness of ULD loss in enabling distillation across models with different architectures and tokenizers, paving the way to a more widespread use of distillation techniques.

1 Introduction

A noticeable trend has emerged in NLP with the prevalence of large language models (LLMs) such as Llama (Touvron et al., 2023a), Mistral (Jiang et al., 2023), Falcon (Almazrouei et al., 2023), GPT-NeoX (Black et al., 2022), or Mixtral (Jiang et al., 2024). Despite the high performance of LLMs (Bubeck et al., 2023), the challenges associated with resource consumption and deployment complexity have become increasingly prominent due to hardware availability, cost, and latency bottlenecks. Several methods, including efficient decoding (Leviathan et al., 2023; Ye et al., 2023), model recycling (Lester et al., 2022), and size reduction (Dettmers et al., 2023; Ma et al., 2023), have been proposed to address the need for faster and more efficient deployment of these models. In this paper, we concentrate on knowledge distillation (KD) (Buciluundefined et al., 2006; Hinton et al., 2015),

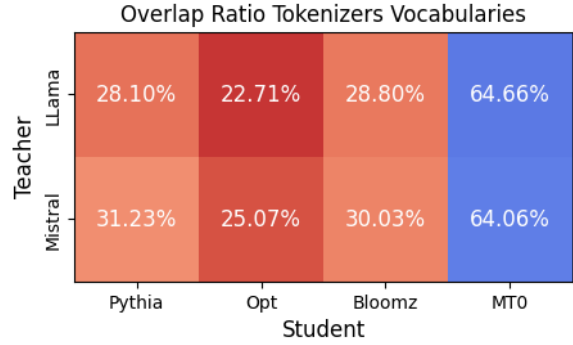


Figure 1: Ratio of student vocabulary contained in teacher tokenizer. (e.g., Bloomz tokenizer has 30.03% of Mistral’s vocabulary). For student and teacher model information see Sec. 4.2

a widely adopted technique (Sanh et al., 2020; Jiao et al., 2020; Mohammadshahi et al., 2022; He et al., 2023; Raman et al., 2023; Dasgupta et al., 2023), utilized by practitioners to distill the expertise of a larger teacher model into a compact student model, preserving maximum performance while reducing latency and memory footprint.

Over the past years, NLP researchers have extensively explored and applied knowledge distillation mostly on smaller student models derived from BERT (Sanh et al., 2020; Jiao et al., 2020; Sun et al., 2020). These smaller student models maintain a similar architecture to the teacher, mirroring some of its blocks, hidden sizes, or relying on the same tokenizer. Two approaches can generally be considered: the *white box* approach, where researchers propose loss functions that directly compute similarities across layers, and the *black box* approach, offering flexibility, agnostic to teacher latent states. This black box approach can be readily implemented by practitioners through libraries and APIs, simplifying its adoption.

However, KD for generative models, those relying on encoder-decoder or decoder architectures, has received less attention. Recent research, pre-

dominantly focuses on synthetic data fine-tuning (He et al., 2022; Kramchaninova and Defauw, 2022; Ouyang et al., 2022), with less emphasis on refining loss functions in the black box approach. Thus far, the primary method for KD involves using text generated by the teacher model (He et al., 2023; Hsieh et al., 2023), and optionally, augmenting it with logits, employing Kullback–Leibler divergence between the teacher logits distribution and student ones (Mohammadshahi et al., 2022; Raman et al., 2023; Wang et al., 2020a). Although the logit distillation method yields significant improvements, it has been underutilized in recent studies due to its requirement for the student model to share the same tokenizer as the teacher (Sec. 2.4 and Fig. 2). Indeed, even when encoder-decoder or decoder-only models of varying sizes are available, they often do not share the same architecture nor the same tokenizer (Fig. 1), rendering Logit Distillation loss inapplicable. This raises the following research question:

How can we craft a Universal knowledge distillation loss within the black box approach, that is inherently versatile to teacher/student architectures and agnostic on tokenizers?

Contributions: In this paper, we make the following contributions:

1. **A universal logit distillation loss.** We introduce a new loss, Universal Logit Distillation Loss (ULD loss), with virtually no assumptions on the teacher and student architectures. Our approach harnesses a closed-form solution of optimal transport, rendering it ideal for large-scale fine-tuning due to its fast computation.
2. **Experimental Results.** To showcase the consistent effectiveness of our loss, we rely on various tasks: extractive question answering, generative question answering, and summarization. Our evaluation spans multiple widely-used teacher-student pairs, with diverse vocabularies and model architectures.
3. **Contributing to future research.** We make our code¹², model weights, and generated datasets³ openly available to facilitate future research, minimizing computational overhead and lowering entry barriers.

¹<https://github.com/Nicolas-BZRD/llm-recipes>

²<https://github.com/Nicolas-BZRD/llm-distillation>

³<https://huggingface.co/collections/Nicolas-BZRD/llms-distillation-65cfa07f1e4ed7404502a9eb>

2 Problem Formulation & Related Work

2.1 Notations

We define Ω a model vocabulary set, $|\Omega|$ his size, and Ω^* signifies its Kleene closure⁴. Let $\mathcal{P}(\Omega)$ denote the probability distributions set defined over the sample space Ω . It is defined as $\mathcal{P}(\Omega) = \mathbf{p} \in [0, 1]^\Omega$ with $\sum_{j=1}^\Omega p_j = 1$. Consider $\mathcal{D}$ as a non-empty set with independent and identically distributed samples $(\mathbf{x}^i)_{i=1}^N \in \Omega^*$. Each $\mathbf{x}^i$ is a sequence of tokens, where $x_j^i \in \Omega$ represents the j th token of the i th sequence. The notation $\mathbf{x}_{<t}^i = (x_{t-1}^i, \dots, x_0^i) \in \Omega^*$ denotes the prefix of length t . In this paper, we will denote by Ω^S the student vocabulary and by Ω^T the teacher’s vocabulary.

Remark. In general $\Omega^T \neq \Omega^S$ and $|\Omega^T| \neq |\Omega^S|$ but $\Omega^T \cap \Omega^S \neq \emptyset$.

Conditional textual generation. Conditional textual generation aims to model the probability distribution $\mathbf{p}_*(\mathbf{x})$ over variable-length text sequence $\mathbf{x}$ by approximating $\mathbf{p}_\theta(\mathbf{x})$ parameterized by $\theta \in \Theta$ to $\mathbf{p}_*(\mathbf{x})$ for any $\mathbf{x}$. In this work, we assume the presence of a teacher (f_{θ_T}) and student (f_{θ_S}), two pre-trained conditional language models with a general form applicable to both $f_\theta : \Omega^* \rightarrow \mathbb{R}^{|\Omega|}$, where the output is provided in unnormalized logit score. The function f_θ parameterizes $\mathbf{p}_\theta$ (f_{θ_T} and f_{θ_S} respectively parameterize $\mathbf{p}_{\theta_S}$ and $\mathbf{q}_{\theta_T}$), i.e., for any sentence $\mathbf{x}$, $\mathbf{p}_\theta(\mathbf{x}) = \text{softmax}\left(\frac{f_\theta(\mathbf{x})}{\tau}\right)$, where $\tau \in \mathbb{R}^+$ denotes the temperature set as default to 1. Given an input sequence $\mathbf{x}$, the pre-trained language model f_θ can iteratively generate an output sequence $\hat{\mathbf{x}}$ during inference by sampling $\hat{x}_{t+1} \sim \mathbf{p}_\theta(\cdot | \hat{\mathbf{x}}_{<t})$ with $\hat{x}_t$ a generate token.

2.2 Knowledge Distillation Framework

In knowledge distillation (KD), the objective is to guide the learning of a student model using a more complex teacher model (Bucilu et al., 2006; Hinton et al., 2015). Generally, this paradigm comprises two key components: a cross-entropy loss ($\mathcal{L}_{CE}$), which guides the student model to predict gold tokens accurately, and a secondary loss tasked to align the probability distributions of the student model with those of the teacher model to facilitate the transfer of general knowledge ($\mathcal{L}_{KD}$).

⁴The Kleene closure corresponds to sequences of arbitrary size written with words in Ω . Formally: $\Omega^* = \bigcup_{i=0}^\infty \Omega^i$.

Formally, the goal of the student is to minimize $\mathcal{L}$:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \times \mathcal{L}_{KD} \quad (1)$$

where $\lambda \in \mathbb{R}^+$ can be used to control the trade-off between learning exclusively from text and knowledge coming from the teacher.

2.3 Knowledge Distillation Related Work

Building on Eq. 1, various cases have been examined.

Distillation from teacher-generated text: Distillation from teacher-generated text occurs when $\lambda = 0$. This strategy is particularly advantageous when dealing with synthetic data (Kramchaninova and Defauw, 2022; Du et al., 2023; Ushio et al., 2023), a fact highlighted by the effectiveness of instructing large language models such as GPT-3.5/4 (Wu et al., 2023; Bubeck et al., 2023). Distillation from teacher-generated text (Kim and Rush, 2016; He et al., 2023; Hsieh et al., 2023; Zhou and Chiam, 2023) will be considered as a baseline throughout the paper. The primary drawback of these methods lies in their failure to fully leverage all the information that can be provided by the teacher.

White-box approach: A further refinement of Eq. 1 occurs when $\mathcal{L}_{KD}$ relies on the internal features of the teacher to transfer knowledge (Jiao et al., 2020; Sun et al., 2020). Popular features include transformer attention and internal layers within both encoder-only and encoder-decoder models (Raman et al., 2023; Wang et al., 2020a, 2021). A main drawback is that these methods demand access to the models’ internal mechanisms, which is typically unavailable through API access. Moreover, they often assume similarities in architectural patterns between the teacher and the student.

Black-box approach: In the black-box approach, practitioners are limited to accessing only the output probabilities or logits of the model. They use these logits to align the student’s output probabilities with those of the teacher through Kullback–Leibler divergence (KL) (Sanh et al., 2020). This method has emerged as one of the most widely adopted approaches, successfully distilling encoder, decoder, or encoder-decoder models (Timiryasov and Tastet, 2023; Mohammadshahi et al., 2022; Zhao et al., 2023a). However, employing KL divergence necessitates that both student and teacher share the same vocabulary, a requirement impractical with current large language models (LLMs) as reported in Fig. 1. We dig into the limitations of this method in the next section.

2.4 KL Distillation loss

When distilling using the KL (Li et al., 2021), the goal is to force the student to learn the teacher’s output probability distribution at each generation step. The formal definition of the objective function is provided in Eq. 2.

$$\mathcal{L} = \sum_{t=1}^{|\mathbf{x}|} \text{CE}(t) + \lambda \text{KL} [\mathbf{q}_{\theta_T}(\cdot|\mathbf{x}_{<t}), \mathbf{p}_{\theta_S}(\cdot|\mathbf{x}_{<t})] \quad (2)$$

where $|\mathbf{x}|$ the length of the tokenized sentence $\mathbf{x}$, $\mathbf{q}_{\theta_T}(\cdot|\mathbf{x}_{<t})$ and $\mathbf{p}_{\theta_S}(\cdot|\mathbf{x}_{<t})$ are the probability distribution for the Teacher and Student models at the t^{th} steps. $\text{CE}(t)$ denotes the cross-entropy loss at each generation time step t , expressed as: $\text{CE}(t) = -\log(\mathbf{p}_{\theta_S}(x_t|\mathbf{x}_{<t}))$ with x_t the gold token for the t^{th} steps and KL the Kullback–Leibler divergence defined as: $\text{KL}[\mathbf{q}_{\theta_T}(\cdot|\mathbf{x}_{<t}), \mathbf{p}_{\theta_S}(\cdot|\mathbf{x}_{<t})] = \sum_{x \in \Omega} \mathbf{q}_{\theta_T}(x|\mathbf{x}_{<t}) \times \log\left(\frac{\mathbf{q}_{\theta_T}(x|\mathbf{x}_{<t})}{\mathbf{p}_{\theta_S}(x|\mathbf{x}_{<t})}\right)$ where $\lambda \in \mathbb{R}^+$ controls the trade-off between the two terms.

Remark. Eq. 2 relies on equality across the vocabulary of the student and the teacher, i.e. $\Omega = \Omega^S = \Omega^T$, enabling us to compute the KL divergence by ensuring similar support of probabilities.

Remark. Eq. 2 also suppose absolute continuity for the distributions $\mathbf{p}_{\theta_S}(\cdot|\mathbf{x}_{<t}) \ll \mathbf{q}_{\theta_T}(\cdot|\mathbf{x}_{<t})$ at each generation step t , making the use of padding impractical.

Our examination of Eq. 2 illustrating in Fig. 2 highlights that both vocabulary and absolute continuity constraints pose challenges to distilling two distinct LLM families with logits using KL loss. In the following section, we introduce our ULD loss, providing a flexible framework for knowledge distillation across a wide range of LLMs.

3 Universal Logit Distillation

3.1 Background on Optimal Transport

Optimal transport mathematically transfers mass between distributions, minimizing cost (Villani et al., 2009; Peyré et al., 2019). In this context, Wasserstein distance, or Earth Mover Distance, robustly measures dissimilarities between distributions. This distance metric has gained popularity in NLP applications such as hallucination detection (Guerreiro et al., 2022; Shuster et al., 2021), clustering (Zhuang et al., 2022; Ye et al., 2017; Xu et al., 2018) or sentence similarity (Colombo et al., 2021; Xu et al., 2018; Bahuleyan et al., 2018).

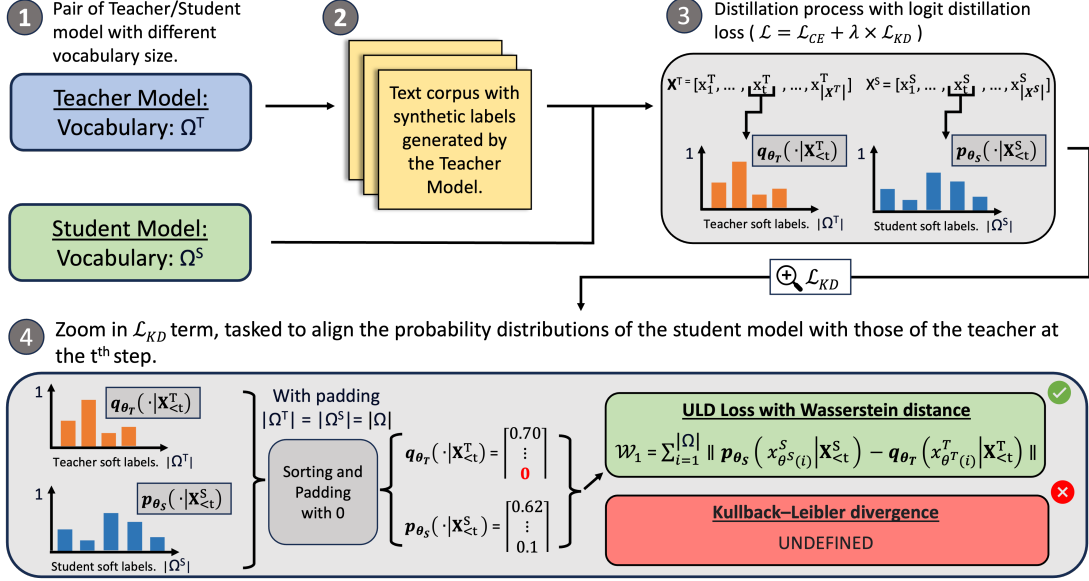


Figure 2: Distillation using ULD loss. In block 4, the KL divergence cannot be defined as the two distributions do not have the same support, breaking the absolute continuity of the quotient in the KL logarithmic term. To alleviate this we rely on the ULD loss which leverages a closed form of the Wasserstein distance.

Wasserstein distance. The Wasserstein distance minimizes transport costs between sampled points from all possible couplings. Let us consider two sets of probability distributions, $\mathcal{P}(\Omega_S)$ and $\mathcal{P}(\Omega_T)$, respectively containing discrete distributions over spaces Ω_S and Ω_T . We denote $\mathbf{p} \in \mathcal{P}(\Omega_S)$ and $\mathbf{q} \in \mathcal{P}(\Omega_T)$ as two discrete probability distributions with $\sum_{i=1}^{|\Omega_S|} p_i \delta_{x_i^S} = 1$ and $\sum_{i=1}^{|\Omega_T|} q_i \delta_{x_i^T} = 1$, where $\delta_{x_i^S}$ and $\delta_{x_i^T}$ represent probability mass points at x_i^S, x_i^T with $x_i^S \in \Omega_S$ and $x_i^T \in \Omega_T$ for distributions $\mathbf{p}$ and $\mathbf{q}$. The values p_i and q_i are weight factors ensuring that the sum of weights is equal to 1. Under this discrete setting, computing the Wasserstein distance is defined as:

$$\mathcal{W}_p(\mathbf{p}, \mathbf{q}) = \min_{T \in \Pi(\mathbf{p}, \mathbf{q})} \sum_{i=1}^{|\Omega_S|} \sum_{j=1}^{|\Omega_T|} T_{ij} C_{ij}^p \quad (3)$$

where $\Pi(\mathbf{p}, \mathbf{q})$ is the set of joint distributions with marginals $\mathbf{p}$ and $\mathbf{q}$, C_{ij}^p represents the cost matrix, and T is the transport plan. For the rest of the work, we focus on the Wasserstein distance related to $p = 1$. Consequently, the Wasserstein distance seeks the optimal approach to transfer probability mass from $\mathbf{p}$ to $\mathbf{q}$, minimizing the transportation cost defined by the absolute norm.

Remark. Note that the Wasserstein distance (see Eq. 3) makes no assumptions about the support of $\mathbf{p}$ or $\mathbf{q}$, unlike the KL divergence, making it a natural choice for distillation.

3.2 Universal Logit Distillation loss

The Universal Logit Distillation loss (ULD loss) is a novel distillation technique designed to virtually distill any generative model teacher into any student. It aims to overcome the limitations of KL divergence, as discussed in Sec. 2.4 and Fig. 2.

Intuition. The ULD loss retains the CE loss term to guide the model in generating the target token and introduces a Wasserstein Distance term to transfer knowledge from the teacher to the student. By minimizing the distance between the soft probabilities of the teacher and the student, our goal is to reproduce not only the predictions for the golden token but also the near-zero labels, which are crucial for performance and generalization.

ULD loss: Formally, the ULD loss function is formulated as:

$$\mathcal{L}_{ULD} = \sum_{t=1}^{|\mathbf{x}|} \text{CE}(t) + \lambda \times \mathcal{W}_1 \left[\mathbf{p}_{\theta_S}(\cdot | \mathbf{x}_{< t}^S), \mathbf{q}_{\theta_T}(\cdot | \mathbf{x}_{< t}^T) \right] \quad (4)$$

where $|\mathbf{x}| = \min(|\mathbf{x}^S|, |\mathbf{x}^T|)$ the minimum length between the sentences tokenized with the teacher or student tokenizers. Respectively $\mathbf{q}_{\theta_T}(\cdot | \mathbf{x}_{< t}^T)$ and $\mathbf{p}_{\theta_S}(\cdot | \mathbf{x}_{< t}^S)$ are the probability distribution for the Teacher and Student models at the t^{th} steps, $\text{CE}(t)$ the cross-entropy loss defined in Sec. 2.4 and $\mathcal{W}_1$ represents the discrete Wasserstein distance defined in Eq. 3 where $\lambda \in \mathbb{R}^+$ controls the trade-off between the two terms and set to 1.5 in the rest of this paper.

Explanation Similar to the KL loss, the discrete Wasserstein distance ensures that the confidence of the student at each time step is close to the one from the teacher.

3.3 Fast Computation & Approximations

To the best of our knowledge, we are the first to motivate and propose the Wasserstein distance as a learning loss for distillation in the scope of the LLM decoder. Prior efforts focus on Sinkhorn regularization (Cuturi, 2013) with encoder-decoder for classification (Bhardwaj et al., 2022), while our focus diverges as we concentrate on the generative setting. This shift presents inherent challenges, as the naive computation of the Wasserstein loss in Eq. 3 exhibits a complexity of $\mathcal{O}(n^3 \log n)$, where n signifies the size of the larger support. While manageable in small classification scenarios with encoders, the magnitude of the vocabulary, which can extend to 100K tokens in generative tasks, renders this approach intractable, particularly for long sequences.

Closed form solution for ULD loss: To achieve efficient computation of the Wasserstein distance in Eq. 4, we introduce two additional refinements:

Uniform Support Length: We augment either the student or teacher vocabulary size through distribution padding (with 0 value), ensuring equal support size for both (i.e., $|\Omega_t| = |\Omega_s| = |\Omega|$).

Uniform Cost: As teacher and student supports differ, and no vocabulary relationship is established, we assert that each transport cost is equal to 1. While this may seem a strong assumption, we will demonstrate that the approximation we draw still achieves better results in our case.

Under this assumption the Wasserstein distance used in the $\mathcal{L}_{\text{ULD}}$ loss becomes (Peyré et al., 2019):

$$\mathcal{W}_1 = \sum_{t=1}^{|\mathbf{x}|} \sum_{i=1}^{|\Omega|} \left| \mathbf{p}(x_{\sigma^S(i)}^S | \mathbf{x}_{<t}^S) - \mathbf{q}(x_{\sigma^T(i)}^T | \mathbf{x}_{<t}^T) \right| \quad (5)$$

where σ^S and σ^T as the permutation that sorts the decreasing probability of student and teacher soft probability vectors.

Intuition. The final version of Eq. 5 is straightforward: with each generation, we pad and sort the probability vectors of both student and teacher, then compute the absolute difference. This ensures that no matter the token, the confidence levels of the sorted teacher and student are close.

4 Experimental Setting

4.1 Evaluation Scenarios

To align with the black-box approach where training models may not be available, we abstain from fine-tuning teacher models. In this way, we enable ULD Loss to operate in an unsupervised environment by generating all *answers* text with teacher models. For repeatability and fair comparison between experiments, we opted to retain original answers for the test set split. We investigated various scenarios to evaluate the ULD loss performance across different datasets and tasks. These comprised 2 Extractive QA (Ext.), 2 Generative QA (Gen.), and 1 Summary (Sum.) tasks:

SQuAD (Ext.): The Stanford Question Answering Dataset (SQuAD) (Rajpurkar et al., 2016) is a reading comprehension dataset with 87,600 questions generated by crowdworkers from Wikipedia articles. Answers are text portions from the relevant sections of the articles.

QED (Ext.): The QED (Lamm et al., 2020) dataset, expertly annotated, extends from a subset of the Google Natural Questions dataset, comprising 7,640 question-answering pairs with explanations. Our focus is exclusively on extracted answers (spans).

FairytaleQA (Gen.): The FairytaleQA Dataset (Xu et al., 2022), created by educational experts, consists of 10,580 questions from 278 children-friendly stories. Questions may be explicit or implicit.

PubMedQA (Gen.): The PubMedQA (Jin et al., 2019) dataset contains question-answer pairs extracted from medical papers. Questions are based on titles, context on abstracts, and responses on conclusions. Due to the dataset size and context length of our student models, we subset the dataset by selecting the first 50,000 smaller items.

DIALOGSum (Sum.): DialogSum (Chen et al., 2021) is a large-scale dialogue summarization dataset, consisting of 13,460 spoken dialogues with corresponding summaries and topics.

4.2 Experimental Choices

Baseline. As far as we know, the only method currently capable of distilling any pair of teacher and student LLM models in a black-box approach is distillation from teacher-generated text seen in [Sec. 2.3](#). Throughout the remainder of this paper, distillation from teacher-generated text will serve as the baseline for evaluating the distillation process using the ULD loss across different teacher-student pairs, tasks, and datasets.

Teacher Models. We employed two teacher decoder models, each with 7 billion parameters: LLama 2 7b Chat (LLama) ([Touvron et al., 2023b](#)) and Mistral 7b Instruct (Mistral) ([Jiang et al., 2023](#)). These instruct models were chosen for their ability to generate few-shot answers ([Brown et al., 2020](#); [Wang et al., 2020b](#)) across diverse tasks and their distinct vocabulary set as shown in [Figure 1](#).

Student Models. We chose student models from various LLM families and architectures with parameters ranging between 160 million to 1 billion: OPT 350m ([Zhang et al., 2022](#)), Pythia 160m, Pythia 410m, Pythia 1b ([Biderman et al., 2023](#)), Bloomz 560m ([Muennighoff et al., 2023](#)) all decoder models and MT0 580m ([Muennighoff et al., 2023](#)) an encoder-decoder. It’s important to note that models can have been already pre-trained on some datasets such as SQuAD for Bloomz and MT0.

Training process. ULD loss distillation and teacher-generated text distillation were processed uniformly. The two teacher models generate answers in inference mode for the five datasets. These answers are then utilized to train student models. During training, student models are trained exclusively to predict answers, in teacher forcing configuration. Logits used for the ULD loss are calculated by applying teacher models to the same data points they generated. Teacher’s weights were frozen, ensuring consistency in teacher-generated sentences during inference and training. Additional parameter details (learning rate, batch size, etc.) can be found in the [Appendix Sec. 11](#).

4.3 Teacher Performances

Distilling using synthetic teacher-generated *answers* might restrict student performance on teacher’s ones. To measure distillation efficiency accurately, we report the average native performances across tasks for both teachers [Tab. 1](#) (details in [Appendix Sec. 9](#)). We chose a primary metric for each task reflecting associate performances: F1

score for Extractive QA ([Sokolova et al., 2006](#)), BERTScore for Generative QA ([Zhang* et al., 2020](#)), and Rouge-Lsum for summary task ([Lin, 2004](#)). Comprehensive evaluation methods and outcomes, encompassing prompts and few-shot examples, are provided in the [Appendix Sec. 10](#).

Model	Extractive (F1)	Generative (BERTScore)	Summary (Rouge-Lsum)
LLama	69.51	36.11	23.90
Mistral	64.66	33.47	34.71

Table 1: **Average performance of teacher models** across tasks with their main metrics. It is important to note a relative difference of 30% in performance between teacher models on the summary task.

5 Empirical Results

5.1 General Results

We empirically validate the effectiveness of the ULD loss step-by-step. First, we report in [Tab. 2](#) the aggregated key metrics performance over the different datasets and teacher/student pairs. **ULD loss achieves the best overall results**, which indicates that the proposed ULD loss effectively improves the performances of every student model on a variety of downstream tasks using any Teacher. Notably, ULD loss exhibits an average improvement of 2.30 points over models trained on teacher-generated text for extractive QA tasks and Bloomz outperforms his teacher Mistral on the QED datasets. Furthermore, concerning summarization tasks, the 30% performance disparity between LLama/Mistral ([Tab. 1](#)) persists in their distilled counterparts ([Tab. 2](#)), underscoring the critical role of teacher performances.

5.2 Student Size Ablation Study

General results in [Tab. 2](#) show a consistent pattern regarding the model size and the gain achieved with the ULD loss, especially for challenging tasks such as generative QA. To understand the impact of student size on distillation capability, we performed an ablation study over the Pythia family. We hold the training dataset size fixed at 100% and compare the performance of models from 160m, 410m to 1b parameters and report results in [Fig. 3](#). We observe that incorporating ULD loss consistently enhances student models across various tasks. The enhancements are particularly noticeable for smaller models on simpler tasks, while ULD loss requires larger

Teacher	Model	Method	SQUAD (F1)	QED (F1)	FairytaleQA (BERTScore)	PubMedQA (BERTScore)	DIALOGSum (Rouge-Lsum)
Teacher	LLama	-	81.30	57.72	41.59	30.62	23.90
	Mistral	-	76.31	53.01	36.01	30.93	34.71
LLama	OPT-350m	Raw Text	70.78	48.64	33.78	27.99	20.58
		ULD Loss	72.97	49.06	33.03	30.01	20.11
	Pythia-410m	Raw Text	71.39	47.04	33.02	29.86	20.94
		ULD Loss	74.14	49.15	34.83	29.89	22.19
	Bloomz-560m	Raw Text	73.54	50.99	36.70	29.14	20.01
		ULD Loss	75.90	55.33	37.86	30.01	22.67
Mistral	OPT-350m	Raw Text	71.64	50.13	30.09	27.91	31.44
		ULD Loss	73.35	50.88	30.44	30.30	32.17
	Pythia-410m	Raw Text	71.50	47.07	31.44	28.25	31.64
		ULD Loss	73.64	50.38	31.79	29.55	33.10
	Bloomz-560m	Raw Text	73.34	52.15	32.64	28.87	31.95
		ULD Loss	76.00	55.79	33.93	30.60	32.58
Average	-	Raw Text	72.03	49.34	32.94	28.67	26.09
	-	ULD Loss	74.33	51.77	33.65	30.06	27.14

Table 2: **Overall performance** of Teacher/Student pair models trained with ULD Loss and teacher-generated text (Raw Text) across tasks with their main metrics. Evaluations are performed over respective test splits.

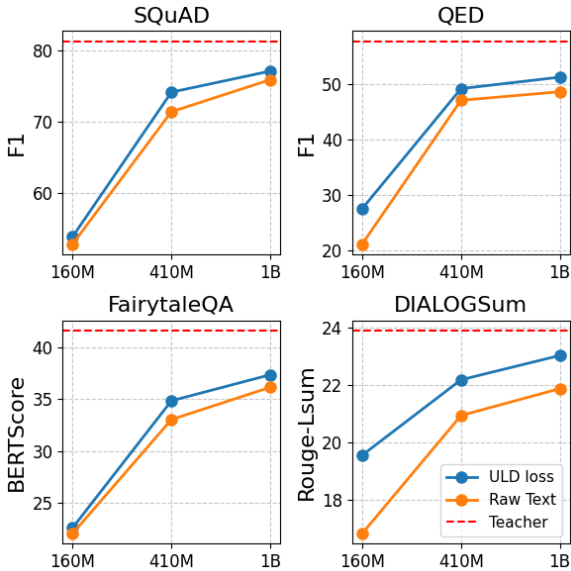


Figure 3: **Student model size ablation** with the Pythia family trained by a LLama teacher. Trainings are conducted with ULD loss and teacher-generated text (raw text). Evaluation scores on test sets are depicted on the Y-axis, while Pythia model sizes are on the X-axis.

models for effectively distilling teacher logits on harder tasks. This is especially evident in tasks requiring reasoning, such as FairytaleQA. While using logits teacher improves training, deep reasoning tasks still require appropriate model sizes to process complex relationships taught by teachers. Generally, we observe a significant increase in capacity transfer from teacher to student models through **the use of ULD, enabling student**

models to match models twice bigger trained with the teacher-generated text method. For example, Pythia 410m with ULD loss matches the performance of the Pythia 1b distilled with teacher-generated text on QED and DIALOGSum.

5.3 Dataset Size Ablation Study

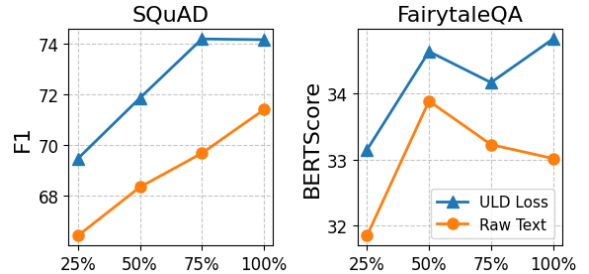


Figure 4: **Dataset size ablation** with a LLama/Pythia-410m pair of models trained with ULD loss or teacher-generated text (raw text). The X-axis indicates the % of data used during training while the y-axis represents the test set score for respective datasets.

In this section, we investigate and report in Fig. 3 the influence of the dataset size for models trained with ULD Loss or teacher-generated text. We perform ablations with respectively 25%, 50%, 75%, and 100% of dataset size while keeping training parameters constants. For every ablation ratio, models trained with ULD loss achieved better performance than models trained on teacher-generated text. Specifically, **with 50% of a dataset, ULD loss models overpass the performance of teacher-**

generated text models trained with full dataset.

5.4 Training Regularization

To understand the impact of the ULD loss during training we decide to compute the validation ULD and Cross-entropy loss values for two pairs of teacher/student on the SQuAD dataset every 200 steps during 5 epochs. We report the curves formed by this point in Fig. 5.

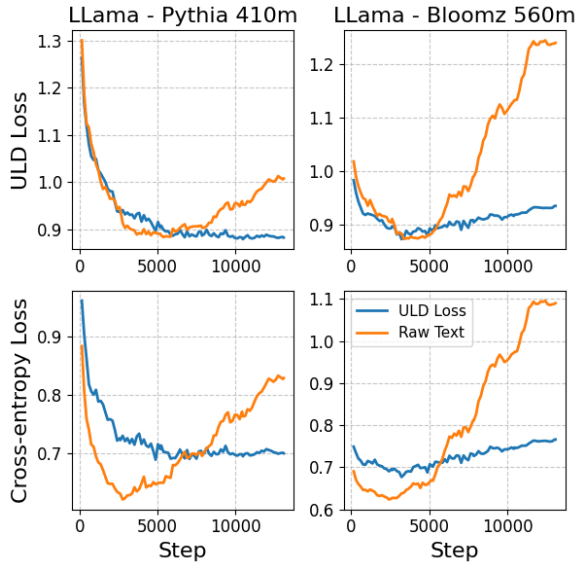


Figure 5: **Evolution of validation ULD and Cross-entropy loss curves during training** on SQuAD dataset for a Llama/Pythia-410m and Llama/Bloomz-560m Teacher/Student pair of model. For teacher-generated text models (raw text), the ULD loss was only computed during validation and did not impact the training process.

It appears that using the **ULD loss contributes to stabilizing the distillation process over training** and mitigates overfitting issues, enabling the model to train more effectively across multiple epochs. It’s worth noting that incorporating the ULD loss during training stabilizes both ULD and Cross-entropy loss.

Remark. Higher Cross-entropy loss validation values observed with ULD loss are a direct result of the training process. ULD loss trained model to predict soft labels from the teacher model instead of the one-hot vectors used for computing the Cross-entropy loss.

6 Distillation of Decoder Teacher to Encoder-Decoder Student

As shown in Sec. 5, ULD loss effectively transfers knowledge from any pair of teacher/student

decoders. Moreover, by leveraging solely on logit information and adopting a black-box approach, ULD loss should be able to extend its versatility and improve cross-architecture distillation. To validate this, we distill a teacher/student pair Llama/MT0-580m and focus our experimentation on PubMedQA, DIALOGSum, and QED to avoid any data seen during the pre-training of MT0 with the xP3 dataset (Muennighoff et al., 2023; Sanh et al., 2022).

ULD loss	QED (F1)	PubMedQA (BERTScore)	DIALOGSum (Rouge-Lsum)
Raw Labels	55.63	27.56	23.22
ULD Loss	56.01	30.19	23.92

Table 3: Distillation of a Llama teacher (decoder) to an MT0-580m (encoder-decoder) with ULD Loss and teacher-generated text on three data sets and their primary metric associate.

The results presented in Tab. 3 demonstrate that incorporating logit information from a decoder teacher using ULD loss can enhance the performance of an encoder-decoder student model. Notably, the inherent ability of the encoder-decoder in the summary task seems to be limited by the synthetic *answers* as teacher-generated text distillation matches the teacher’s performance. However, by using the logit information with the ULD loss, the student model still leads to improved results, suggesting a successful knowledge transfer through logits. With this additional knowledge, the student model slightly outperforms the teacher one. Furthermore, in generative tasks where decoder architectures perform, the encoder-decoder student model gained 2.63 points over distillation with teacher-generated text.

7 Conclusions

In this work, we introduce the Universal Logit Distillation (ULD) loss, a novel method for distilling any decoder teacher model into any student model for LLM generative tasks. ULD achieves better overall results and matches the performance of teacher-generated text distillation with only half of the training dataset or student size, while effectively preventing overfitting. Our extensive experiments validate the efficacy of the ULD loss across diverse tasks,- datasets, and architectures, demonstrating its superiority over standard teacher-generated text distillation methods.

Limitations

One unexplored avenue related to our work is the study of the ULD loss in non-English languages, including ones with vastly different token representations (Chinese, Korean, etc). However, we believe that our findings can still be applied to other languages, especially if teacher-student model pairs are pre-trained in the same language or achieve multilingual capacity. Additionally, in line with previous work, we evaluate task performance with standard reference-based metrics (ROUGE, F1, BERTScore) which can be limited in their assessment of what constitutes a correct generative model prediction (Faysse et al., 2023). An extension of the work could be to go beyond mono-task distillation, and assess the transferability of generalist assistant abilities from larger to smaller models.

Ethics Statement

Knowledge distillation aims to reduce the size, cost, and energy consumed by a model at inference time. Our work opens new perspectives in this area, aligned with the desire for sobriety, notably for environmental reasons. Although KD allows partial transfer of larger model performance, smaller models remain limited in their reasoning capacity and are more susceptible to hallucinatory behavior (Rawte et al., 2023), especially in open-ended generation tasks. This phenomenon has not been extensively studied in this work. Furthermore, by distilling knowledge from existing models, if a bias is already present in the teacher model, it may be transferred to the student model. This is not unique to our method, but it’s a common risk for all knowledge distillation.

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8 Appendix - General Results

8.1 Summary

Teacher	Model	Method	Dataset	Rouge-1	Rouge-2	Rouge-L	Rouge-Lsum
Llama	Bloomz-560m	Raw Text	DIALOGSum	24.71	10.06	19.99	20.01
Llama	Bloomz-560m	ULD Loss	DIALOGSum	28.08	11.68	22.64	22.67
Mistral	Bloomz-560m	Raw Text	DIALOGSum	39.85	15.36	31.92	31.95
Mistral	Bloomz-560m	ULD Loss	DIALOGSum	40.57	15.94	32.6	32.58
Llama	OPT-350m	Raw Text	DIALOGSum	25.4	10.48	20.57	20.58
Llama	OPT-350m	ULD Loss	DIALOGSum	23.69	9.76	20.13	20.11
Mistral	OPT-350m	Raw Text	DIALOGSum	39.33	14.97	31.49	31.44
Mistral	OPT-350m	ULD Loss	DIALOGSum	39.8	15.76	32.19	32.17
Llama	Pythia-410m	Raw Text	DIALOGSum	26.28	10.52	20.92	20.94
Llama	Pythia-410m	ULD Loss	DIALOGSum	27.29	11.2	22.17	22.19
Mistral	Pythia-410m	Raw Text	DIALOGSum	39.69	15.0	31.62	31.64
Mistral	Pythia-410m	ULD Loss	DIALOGSum	41.39	15.93	33.08	33.1
Llama	Pythia-160m	Raw Text	DIALOGSum	20.34	7.46	16.81	16.81
Llama	Pythia-160m	ULD Loss	DIALOGSum	22.94	8.39	19.56	19.55
Llama	Pythia-1b	Raw Text	DIALOGSum	27.71	11.08	21.86	21.88
Llama	Pythia-1b	ULD Loss	DIALOGSum	28.48	12.16	23.04	23.04

Table 4: Details performance of Teacher/Student pair models trained with ULD Loss and teacher-generated text (Raw Text) for the Summary task. Evaluations are performed over respective test splits.

8.2 Extractive QA

Teacher	Model	Method	Dataset	F1	Precision	Recall
Llama	Bloomz-560m	Raw Text	SQuAD	73.54	75.35	75.19
Llama	Bloomz-560m	ULD Loss	SQuAD	75.9	77.37	77.88
Llama	Bloomz-560m	Raw Text	QED	50.99	58.9	52.38
Llama	Bloomz-560m	ULD Loss	QED	55.33	63.22	56.47
Mistral	Bloomz-560m	Raw Text	SQuAD	73.34	73.31	78.52
Mistral	Bloomz-560m	ULD Loss	SQuAD	76.0	76.1	81.1
Mistral	Bloomz-560m	Raw Text	QED	52.15	57.49	56.28
Mistral	Bloomz-560m	ULD Loss	QED	55.79	61.98	58.8
Llama	OPT-350m	Raw Text	SQuAD	70.78	72.52	72.78
Llama	OPT-350m	ULD Loss	SQuAD	72.97	74.61	74.99
Llama	OPT-350m	Raw Text	QED	48.64	54.74	51.84
Llama	OPT-350m	ULD Loss	QED	49.06	55.38	51.74
Mistral	OPT-350m	Raw Text	SQuAD	71.64	71.67	77.28
Mistral	OPT-350m	ULD Loss	SQuAD	73.35	73.25	78.91
Mistral	OPT-350m	Raw Text	QED	50.13	55.36	54.56
Mistral	OPT-350m	ULD Loss	QED	50.88	56.61	54.53
Llama	Pythia-410m	Raw Text	SQuAD	71.39	73.76	72.85
Llama	Pythia-410m	ULD Loss	SQuAD	74.14	75.88	76.31
Llama	Pythia-410m	Raw Text	QED	47.04	54.31	48.87
Llama	Pythia-410m	ULD Loss	QED	49.15	54.75	53.13
Mistral	Pythia-410m	Raw Text	SQuAD	71.5	71.33	77.54
Mistral	Pythia-410m	ULD Loss	SQuAD	73.64	73.34	79.71
Mistral	Pythia-410m	Raw Text	QED	47.07	50.2	54.67
Mistral	Pythia-410m	ULD Loss	QED	50.38	54.19	56.62
Llama	Pythia-160m	Raw Text	SQuAD	52.83	53.68	56.67
Llama	Pythia-160m	ULD Loss	SQuAD	53.86	54.57	58.19
Llama	Pythia-160m	Raw Text	QED	21.11	21.69	34.46
Llama	Pythia-160m	ULD Loss	QED	27.48	30.3	33.8
Llama	Pythia-1b	Raw Text	SQuAD	75.89	77.36	78.28
Llama	Pythia-1b	ULD Loss	SQuAD	77.1	78.57	79.55
Llama	Pythia-1b	Raw Text	QED	48.59	51.05	60.41
Llama	Pythia-1b	ULD Loss	QED	51.22	55.3	59.7

Table 5: Details performance of Teacher/Student pair models trained with ULD Loss and teacher-generated text (Raw Text) for Extractive QA task. Evaluations are performed over respective test splits.

8.3 Generative QA

Teacher	Model	Method	Dataset	BERTScore	PBert	RBert
Llama	Bloomz-560m	Raw Text	FairytaleQA	36.7	45.42	28.46
Llama	Bloomz-560m	ULD Loss	FairytaleQA	37.86	46.93	29.36
Llama	Bloomz-560m	Raw Text	PubMedQA	29.14	29.45	28.86
Llama	Bloomz-560m	ULD Loss	PubMedQA	30.01	32.5	27.65
Mistral	Bloomz-560m	Raw Text	FairytaleQA	32.64	39.46	26.09
Mistral	Bloomz-560m	ULD Loss	FairytaleQA	33.93	42.45	25.8
Mistral	Bloomz-560m	Raw Text	PubMedQA	28.87	28.59	29.14
Mistral	Bloomz-560m	ULD Loss	PubMedQA	30.6	32.08	29.13
Llama	OPT-350m	Raw Text	FairytaleQA	33.78	41.46	26.57
Llama	OPT-350m	ULD Loss	FairytaleQA	33.03	41.16	25.32
Llama	OPT-350m	Raw Text	PubMedQA	27.99	28.46	27.56
Llama	OPT-350m	ULD Loss	PubMedQA	30.01	36.11	24.14
Mistral	OPT-350m	Raw Text	FairytaleQA	30.09	35.47	24.91
Mistral	OPT-350m	ULD Loss	FairytaleQA	30.44	37.38	23.81
Mistral	OPT-350m	Raw Text	PubMedQA	27.91	27.06	28.75
Mistral	OPT-350m	ULD Loss	PubMedQA	30.3	36.99	23.82
Llama	Pythia-410m	Raw Text	FairytaleQA	33.02	41.31	25.26
Llama	Pythia-410m	ULD Loss	FairytaleQA	34.83	42.61	27.49
Llama	Pythia-410m	Raw Text	PubMedQA	29.86	31.06	28.72
Llama	Pythia-410m	ULD Loss	PubMedQA	29.89	31.23	28.62
Mistral	Pythia-410m	Raw Text	FairytaleQA	31.44	37.97	25.18
Mistral	Pythia-410m	ULD Loss	FairytaleQA	31.79	38.17	25.71
Mistral	Pythia-410m	Raw Text	PubMedQA	28.25	25.91	30.56
Mistral	Pythia-410m	ULD Loss	PubMedQA	29.55	30.09	29.01
Llama	Pythia-160m	Raw Text	FairytaleQA	22.03	30.05	14.5
Llama	Pythia-160m	ULD Loss	FairytaleQA	22.58	31.61	14.08
Llama	Pythia-160m	Raw Text	PubMedQA	26.54	26.26	26.85
Llama	Pythia-160m	ULD Loss	PubMedQA	29.78	36.4	23.36
Llama	Pythia-1b	Raw Text	FairytaleQA	36.13	46.11	26.74
Llama	Pythia-1b	ULD Loss	FairytaleQA	37.34	46.93	28.33
Llama	Pythia-1b	Raw Text	PubMedQA	30.12	31.67	28.64
Llama	Pythia-1b	ULD Loss	PubMedQA	29.88	30.4	29.44

Table 6: Details performance of Teacher/Student pair models trained with ULD Loss and teacher-generated text (Raw Text) for generative tasks. Evaluations are performed over respective test splits.

9 Appendix - Native Performances

Base models used as teacher and student can be respectively download on HuggingFace: Llama 2 7b Chat⁵, Mistral 7b Instruct⁶, Pythia 160m, Pythia 410m, Pythia 1b⁷, Bloomz 560m, MT0 580m⁸.

9.1 Summary

Model	Dataset	Number Few-Shot	Few-Shot Titled	Rouge-1	Rouge-2	Rouge-L	Rouge-Lsum
Bloomz-560m	DIALOGSum	3	False	15.36	1.47	11.92	11.9
OPT-350m	DIALOGSum	2	False	22.06	3.31	17.86	17.84
Pythia-410m	DIALOGSum	3	False	23.38	6.4	20.12	20.13
Pythia-160m	DIALOGSum	3	False	16.0	4.72	13.88	13.88
Pythia-1b	DIALOGSum	3	False	33.95	11.85	29.06	29.1
Llama	DIALOGSum	3	False	0.3	0.13	0.24	0.24
Mistral	DIALOGSum	2	False	0.43	0.18	0.35	0.35

Table 7: Native performance details of Teacher/Student pair models benchmark in few-shot setting for the Summary task. Evaluations are performed over respective test splits.

9.2 Extractive QA

Model	Dataset	Number Few-Shot	Few-Shot Titled	F1	Precision	Recall
Bloomz-560m	SQuAD	3	False	66.05	68.6	66.09
Bloomz-560m	QED	3	False	41.01	51.55	38.83
OPT-350m	SQuAD	3	False	30.01	29.34	41.04
OPT-350m	QED	3	False	30.21	32.82	37.6
Pythia-410m	SQuAD	3	False	37.4	36.58	47.55
Pythia-410m	QED	3	False	33.35	38.05	37.02
Pythia-160m	SQuAD	3	False	15.05	16.39	18.83
Pythia-160m	QED	3	False	15.48	20.15	17.31
Pythia-1b	SQuAD	3	False	48.41	48.52	55.55
Pythia-1b	QED	3	False	41.72	47.18	45.76
Llama	SQuAD	1	False	0.81	0.83	0.84
Llama	QED	5	False	0.58	0.64	0.63
Mistral	SQuAD	3	True	0.76	0.74	0.89
Mistral	QED	5	True	0.53	0.55	0.68

Table 8: Native performance details of Teacher/Student pair models benchmark in few-shot setting for extractive QA tasks. Evaluations are performed over respective test splits.

9.3 Generative QA

Model	Dataset	Number Few-Shot	Few-Shot Titled	BERTScore	PBert	RBert
Bloomz-560m	FairytaleQA	3	False	27.43	31.42	23.67
Bloomz-560m	PubMedQA	3	False	-20.3	-9.43	-30.97
OPT-350m	FairytaleQA	3	False	3.82	-4.33	13.1
OPT-350m	PubMedQA	3	False	19.98	23.29	16.94
Pythia-410m	FairytaleQA	3	False	6.76	1.82	12.43
Pythia-410m	PubMedQA	3	False	25.65	30.13	21.42
Pythia-160m	FairytaleQA	3	False	-0.96	-6.87	6.3
Pythia-160m	PubMedQA	3	False	21.35	27.14	15.93
Pythia-1b	FairytaleQA	3	False	20.59	22.4	19.23
Pythia-1b	PubMedQA	3	False	26.13	29.29	23.24
Llama	FairytaleQA	2	False	0.42	0.48	0.36
Llama	PubMedQA	3	False	0.31	0.3	0.32
Mistral	FairytaleQA	5	True	0.41	0.38	0.45
Mistral	PubMedQA	3	False	0.31	0.28	0.34

Table 9: Native performance details of Teacher/Student pair models benchmark in few-shot setting for generative QA tasks. Evaluations are performed over respective test splits.

⁵<https://huggingface.co/meta-llama/Llama-2-7b-chat-hf>

⁶<https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2>

⁷<https://huggingface.co/EleutherAI>

⁸<https://huggingface.co/bigscience>

10 Appendix - Few-Shot examples and Prompt Systems

The few-shot technique was used to generate synthetic data with the teacher. The number of few-shots reported for evaluating teacher models in [Sec. 9](#) are the same numbers used to generate the synthetic answers. It's also important to note that the few-shot method was only used to determine the native performance of the teacher and student, not the distilled versions.

10.1 Prompt Systems

List of prompt system used with teacher templates. The default templates for chat models provided with the huggingface tokenizer have been retained:

- **Extractive QA:** You are an agent answering questions as part of a reading comprehension activity. You must read and understand the context text step by step. Answers are brief and consist exclusively of continuous words taken from the context text provided.
- **Generative QA:** You are an expert agent in reading comprehension (question answering). You must read and understand the contextual text step by step, then answer the question. The answer must be brief.
- **Summary:** You're an expert at summarizing dialogues. You have to read the dialogue between two people and summarize it in no more than one sentence. The summary should be as short as possible, not re-explaining the dialogue in detail and using the person's name when implicitly mentioned.

10.2 Few-Shot examples

Title	Context	Question	Answer
Christine's boyfriend	Patrick Harris (Tim DeKay), Old Christine's new boyfriend, who she meets in a video store and starts dating.	Who played patrick on new adventures of old christine?	Tim DeKay
June 14, 2018: Death Row Inmates	As of June 14, 2018, there were 2,718 death row inmates in the United States.	Total number of death row inmates in the us?	2,718
Modern Communism	Most modern forms of communism are grounded at least nominally in Marxism, an ideology conceived by noted sociologist Karl Marx during the mid nineteenth century.	Who came up with the idea of communism?	Karl Marx
Napoleon's Defeat by Seventh Coalition	A French army under the command of Napoleon Bonaparte was defeated by two of the armies of the Seventh Coalition : a British-led Allied army under the command of the Duke of Wellington, and a Prussian army under the command of Gebhard Leberecht von Blücher, Prince of Wahlstatt.	Who commanded british forces at the battle of waterloo?	The Duke of Wellington
Canine character	Astro is a canine character on the Hanna-Barbera cartoon, The Jetsons.	What was the dog's name on the jetsons?	Astro

Table 10: Few-shot examples for extractive QA used to benchmark models and generate synthetic answers from teachers.

Context	Summary
#Person1#: John, shall we go to Sun Store? I have decided to buy that Murrberry handbag. Anyway, I'm not carrying this one to Mary's wedding. #Person2#: But, Jane, why not rent one with Handbag Hire? Instead of 990, pay 50, and you have it for a whole week. #Person1#: Sounds great, but I never knew I can rent a handbag. #Person2#: Handbag Hire is a new business. It was founded two months ago. Its collection covers many designer handbags. #Person1#: So... for the price of one Murrberry, I can use a different bag each week for twenty weeks? #Person2#: Absolutely. And if you like one of them, you can choose to buy it at a discounted rate. Of course, the price varies by age and condition. For example, a \$ 1500 Murrberry bag can sell for just \$750. #Person1#: Great, but how do I rent? By telephone? Or in person? #Person2#: Either. And more conveniently, it accepts online orders. #Person1#: I'll do it online now. I still have one more question. Mary's wedding is next Saturday. There are only five days left. Do I have enough time? #Person2#: Don't worry. It promises that customers receive their orders by post within two days. Three more days to go. #Person1#: Oh, I'd better order one right now.	Jane wants to buy that Murrberry handbag to carry to Mary's wedding, but John suggests renting one with Handbag Hire and tells her about the service in detail. Jane is pleased to have a try.
#Person1#: The summers are so great here! Not hot at all. I love the cooling breezes, the clear air, all the greenery. #Person2#: This really has been a wonderful holiday for us. Shall we take a walk around the pond or into those woods for a while? #Person1#: Let's do both! Are we in a rush or anything? #Person2#: No, not really. I had thought we'd stay in Hamburg tonight, but we can't unless we rush it. Let's stay in Bremen instead. Tomorrow we can have lunch in Hamburg, then check into a hostel in Copenhagen and have dinner there. #Person1#: Sounds fine to me. Whatever, let's enjoy this pond first. #Person2#: Sure. We can walk around to that path that leads into the woods there. Hey, look! There are some wild ducks over there in the reeds. #Person1#: I see them! Wow! How do you know they're wild? #Person2#: I used to go hunting with my uncle, that's how. #Person1#: They're neat. Now let's take that path into the woods and see what we can see...	#Person1# and #Person2# are enjoying a pond. #Person1# and #Person2# had planned to stay in Hamburg tonight, but they decide to stay in Bremen since they are not in a rush.
#Person1#: Well, Rebecca, is there anything else you need to know for now? #Person2#: I don't think so, Mr. Parsons. I think you have covered all the main points for me. #Person1#: Okay well listen, here is my business card with my mobile number. If any other questions spring to mind don't hesitate to contact me. Of course, you can also call Miss Childs too. #Person2#: Great. Rmm, when can I expect to hear from you? #Person1#: Well, we are finishing the shortlist interviews tomorrow, so we will certainly have a decision made by early next week. Miss Childs will call you to discuss more on Monday or Tuesday. How does that sound? #Person2#: That sounds perfect. Thank you very much for taking the time to speak to me Mr. Parsons. #Person1#: The pleasure's all mine, Rebecca. #Person2#: I hope to hear from you very soon. #Person1#: Absolutely. Thanks for coming Rebecca. Goodbye.	Mr. Parsons gives Rebecca his business card after the interview and tells Rebecca the decision will be made by early next week and Miss Childs will contact Rebecca.

Table 11: Few-shot examples for summary used to benchmark models and generate synthetic summary from teachers.

Title	Context	Question	Answer
The Wee Bannock	So, she jumped up with her lint and her lint cards, and the tailor jumped up with his great shears, and one apprentice grasped the line measure, while another took up the saucer full of pins; and they all tried to catch the wee bannock. But it dodged them round and round the fire, and at last it got safely out of the door and ran down the road, with one of the apprentices after it, who tried to snip it in two with his shears. It ran too quickly for him, however, and at last he stopped and went back to the house, while the wee bannock ran on until it came to a tiny cottage by the roadside. it trundled in at the door, and there was a weaver sitting at his loom, with his wife beside him, winding a clue of yarn.	How did the bannock escape from the tailor's wife and the three tailors?	Dodged them round and round the fire, and at last it got safely out of the door and ran down the road.
Princess Mountain	Then he took the prince by the hand, led him deep down in the earth into his cave, and there on the wall hung a suit of armor altogether forged of the clearest silver, and so bright that it shone afar. Right beside it stood a snow-white steed, saddled and bridled, pawing the earth with his silver hoofs, and champing his bit till the foam dropped to the ground. The wild man said: 'now get quickly into your armor, ride out and try your luck! in the meantime I will tend your oxen.' The prince did not wait to be told a second time; but put on his helmet and armor in all haste, securely buckled on his spurs, hung his sword at his side, and felt as light in his silver armor as a bird in the air. Then he leaped into the saddle so that every clasp and buckle rang, laid his reins on the neck of his steed, and rode hastily toward the glass mountain.	What was the suit of armor given by the wild man forged from?	The clearest silver.
Money Box	He knew very well that he had enough inside him to buy up all the other toys, and this gave him a very good opinion of his own value. The rest thought of this fact also, although they did not express it, for there were so many other things to talk about. A large doll, still handsome, though rather old, for her neck had been mended, lay inside one of the drawers which was partly open. She called out to the others, 'let us have a game at being men and women, that is something worth playing at.'	Why didn't the other toys talk about how valuable the pig was?	There were so many other things to talk about.
A Legend of Confucius	When confucius came to the earth, the kilin, that strange beast which is the prince of all four-footed animals, and only appears when there is a great man on earth, sought the child and spat out a jade whereon was written: 'son of the watercrystal you are destined to become an uncrowned king!' and confucius grew up, studied diligently, learned wisdom and came to be a saint. He did much good on earth, and ever since his death has been revered as the greatest of teachers and masters. He had foreknowledge of many things and even after he had died, he gave evidence of this.	Why was confucius's death revered as the greatest of teachers and masters?	He did much good on earth.
Naughty Boy	'Oh, let me in! Let me in! I'm cold, and I'm so wet!' Exclaimed suddenly a child that stood crying at the door and knocking for admittance, while the rain poured down, and the wind made all the windows rattle. 'Poor thing!' said the old poet, as he went to open the door. there stood a little boy, quite naked, and the water ran down from his long golden hair. He trembled with cold, and had he not come into a warm room he would most certainly have perished in the frightful tempest.	Why did the boy ask to come inside?	He was cold and wet.

Table 12: Few-shot examples for generative QA used to benchmark models and generate synthetic answers from teachers.

Context	Question	Answer
Injury severity score (ISS), Glasgow coma score (GCS), and revised trauma score (RTS) are the most frequently used methods to evaluate the severity of injury in blunt trauma patients. ISS is too complicated to assess easily and GCS and RTS are easy to assess but somewhat subjective. White blood cell count (WBC) is an easy, quick and objective test. This study was performed to evaluate the significance of the WBC count at presentation in the blunt trauma patients. 713 blunt trauma patients, who were admitted to the Uludag University Medical Center Emergency Department between 01.04.2000-31.12.2000, were retrospectively evaluated in terms of ISS, GCS, RTS and white blood cell count at presentation. Statistical analysis revealed that WBC was correlated positively with ISS, but negatively with GCS and RTS.	Does the leukocyte count correlate with the severity of injury	The leukocyte count at presentation can be used as an adjunct in the evaluation of the severity of injury in blunt trauma patients.
The aim of this study was to assess the diagnostic value of articular sounds, standardized clinical examination, and standardized articular ultrasound in the detection of internal derangements of the temporomandibular joint. Forty patients and 20 asymptomatic volunteers underwent a standardized interview, physical examination, and static and dynamic articular ultrasound. Sensitivity, specificity, and predictive values were calculated using magnetic resonance as the reference test. A total of 120 temporomandibular joints were examined. Based on our findings, the presence of articular sounds and physical signs are often insufficient to detect disk displacement. Imaging by static and dynamic high-resolution ultrasound demonstrates considerably lower sensitivity when compared with magnetic resonance. Some of the technical difficulties resulted from a limited access because of the presence of surrounding bone structures.	Internal derangement of the temporomandibular joint: is there still a place for ultrasound?	The present study does not support the recommendation of ultrasound as a conclusive diagnostic tool for internal derangements of the temporomandibular joint.
Figures from the British Defence Dental Services reveal that serving personnel in the British Army have a persistently lower level of dental fitness than those in the Royal Navy or the Royal Air Force. No research had been undertaken to ascertain if this reflects the oral health of recruits joining each Service. This study aimed to pilot a process for collecting dental and sociodemographic data from new recruits to each Service and examine the null hypothesis that no differences in dental health existed. Diagnostic criteria were developed, a sample size calculated and data collected at the initial training establishments of each Service. Data for 432 participants were entered into the analysis. Recruits in the Army sample had a significantly greater prevalence of dental decay and greater treatment resource need than either of the other two Services. Army recruits had a mean number of 2.59 (2.08, 3.09) decayed teeth per recruit, compared to 1.93 (1.49, 2.39 p<0.01) in Royal Navy recruits and 1.26 (0.98, 1.53 p<0.001) in Royal Air Force recruits. Among Army recruits 62.7% were from the two most deprived quintiles of the Index of Multiple Deprivation compared to 42.5% of Royal Naval recruits and 36.6% of Royal Air Force recruits.	Is there a differential in the dental health of new recruits to the British Armed Forces?	A significant difference in dental health between recruits to each Service does exist and is a likely to be a reflection of the sociodemographic background from which they are drawn.

Table 13: Few-shot examples for generative QA used to benchmark models and generate synthetic answers from teachers for medical topic.

11 Appendix - Training Information

During training, all distillation processes were performed over 5 epochs with a batch size of 8 for the SQuAD dataset and 4 for the others. A one-cycle learning rate scheduler was used with the following configuration for decoder models: $\text{max_lr} = 1e - 6$, $\text{initial_lr} = \text{max_lr}/2$, $\text{min_lr} = \text{initial_lr}/5$. For mt0 (encoder-decoder), the max learning rate parameter varied according to datasets: DIALOGSum: $1e-4$, PubMedQA: $3e-4$, and QED: $7e-6$. Finally, distillation was performed in BFLOAT16 mode ([Kalamkar et al., 2019](#)), on 4*NVIDIA A100-SXM4-80GB with the Fully Sharded Data Parallel (FSDP) technique ([Zhao et al., 2023b](#)). In total 4.923 GPU hours were used (i.e. consumption for the entire project in tonnes of CO₂: 0.268).